From Univalent Type Theory to Bishop Set Theory

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Our aim is

to describe how one can translate notions and results from Univalent Type Theory (UTT) to Bishop's theory of sets (BST).

From BST to MLTT, part I

1. A Bishop set is interpreted as a setoid and a function between Bishop sets is interpreted as a setoid-function (e.g., in Chapter 3 of [6] "Bishop's set theory in type theory", and in [7]).
2. This translation is not a faithful interpretation in the sense of Feferman: if $A : \mathcal{U}$, then $=_A$ is the least reflexive relation on A (J-rule) and the free setoid on A is $\varepsilon A := (A, =_A)$, which is a choice setoid, and hence a projective setoid. Since every setoid is the quotient of the projective setoid εA, the presentation axiom follows.
3. Constructive analysis in MLTT (see the Nuprl-book [5], and [6], Chapter 5, Palmgren wanted to translate the whole of [4] in MLTT).
4. Constructive algebra in MLTT: Chapter 4 of [6], a type theoretic reading of [9].

From BST to MLTT, part II

1. In book-HoTT, see section 11.5 “Compactness of the interval”. A definition of completion of a metric space due to Richman (Lawvere) is also included (Ex.11.9)
2. Escardó: “Various new theorems in constructive mathematics developed in a constructive univalent type theory using Agda”.
3. From naive constructive Category Theory in [9] to Category Theory in book-HoTT (with many intermediate approachess).

   Palmgren [10]: “the notion of univalent category is too restrictive to cover many familiar examples”.

The standard neutrality of BISH

BISH is in the "common" territory between $\mathrm{CLASS}, \mathrm{INT}, \mathrm{RUSS}$.

The flexibility of BISH w.r.t. proof-relevance

- W.r.t. proof-relevance, BISH is between the proof-irrelevant CLASS and the universally proof-relevant MLTT (and its extensions).
- The proof-relevance of BISH can be put under the carpet (what Bishop did on purpose), and it can be highlighted, if one chooses to do so (see Feferman [8], Beeson [1], [2], and Schwichtenberg [15]).
- In this way BISH can be seen as a more flexible system of constructive mathematics than MLTT and its extensions (See Deligne's talk at the memorial meeting of Voevodsky).
- Proof-relevance within BISH is based on appropriate definitions, while proof-relevance in MLTT and its extensions is generated "universally" and "automatically".

Proof-relevance and the axiom of choice

- CLASS: the AC as the result of proof-irrelevance.
- The use of some choice (CC, DC) in BISH is an indication of missing witnessing data.
- The $\prod$- and $\sum$-sets of a set-indexed family of Bishop sets satisfy the type theoretic AC (the distributivity of $\prod$ over $\sum$ is shown in [12]).
- AC is provable in MLTT.
- The univalent axiom of choice (UAC), formulated with the use of truncation that suppresses proof-relevance, is not provable in MLTT.

Choice and missing witnessing data

$$\mathrm{Conv}_x\big((x_n)_{n\in\mathbb{N}}\big) :\Leftrightarrow \exists_{C\in\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)}\big(C : x_n \xrightarrow{n} x\big)$$

$$C : x_n \xrightarrow{n} x :\Leftrightarrow \forall_{k\in\mathbb{N}^+}\forall_{n\geq C(k)}\left(|x_n - x| \leq \frac{1}{k}\right)$$

$$\mathtt{PrfMemb}_0^{\mathrm{Conv}_x}\big((x_n)_{n\in\mathbb{N}}\big) := \big\{C \in \mathbb{F}(\mathbb{N}^+,\mathbb{N}^+) \mid C : x_n \xrightarrow{n} x\big)\big\}$$

$$\mathrm{Conv}\big((x_n)_{n\in\mathbb{N}}\big) :\Leftrightarrow \exists_{x\in\mathbb{R}}\exists_{C\in\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)}\big(C : x_n \xrightarrow{n} x\big)$$

$$\mathtt{PrfMemb}_0^{\mathrm{Conv}}\big((x_n)_{n\in\mathbb{N}}\big) := \big\{(x,C) \in \mathbb{R}\times\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+) \mid C : x_n \xrightarrow{n} x\big)\big\}$$

$$\mathrm{Cauchy}\big((x_n)_{n\in\mathbb{N}}\big) :\Leftrightarrow \exists_{C\in\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)}\big(C : \mathrm{Cauchy}\big((x_n)_{n\in\mathbb{N}}\big),$$

$$C : \mathrm{Cauchy}\big((x_n)_{n\in\mathbb{N}}\big) :\Leftrightarrow \forall_{k\in\mathbb{N}^+}\forall_{n,m\geq C(k)}\left(|x_n - x_m| \leq \frac{1}{k}\right)$$

$$\mathtt{PrfMemb}_0^{\mathrm{Cauchy}}\big((x_n)_{n\in\mathbb{N}}\big) := \big\{C \in \mathbb{F}(\mathbb{N}^+,\mathbb{N}^+) \mid C : \mathrm{Cauchy}\big((x_n)_{n\in\mathbb{N}}\big)\big\}$$

The proof-set of an implication

Proposition

If $(x_n)_{n\in\mathbb{N}^+} \in \mathbb{F}(\mathbb{N}^+, \mathbb{R})$ *and* $x_0 \in \mathbb{R}$, *then*

$$x_n \xrightarrow{n} x_0 \Rightarrow (x_n)_{n\in\mathbb{N}^+} \text{ is Cauchy}$$

$$\chi :\Leftrightarrow \phi(x_n, x_0) \Rightarrow \psi(x_n)$$

Proof.

If $C : x_n \xrightarrow{n} x_0$, then $D : \mathrm{Cauchy}\big((x_n)_{n\in\mathbb{N}^+}\big)$, where $D(k) := C(2k)$, for every $k \in \mathbb{N}^+$. □

$$\mathtt{Prf}(\chi) := \mathbb{F}\left(\mathtt{PrfMemb}_0^{\mathrm{Conv}_{x_0}}\big((x_n)_{n\in\mathbb{N}^+}\big), \mathtt{PrfMemb}_0^{\mathrm{Cauchy}}\big((x_n)_{n\in\mathbb{N}^+}\big)\right)$$

Instance of the BHK-interpretation of BISH within BST.

The "converse" implication

Proposition

If $(x_n)_{n\in\mathbb{N}^+} \in \mathbb{F}(\mathbb{N}^+,\mathbb{R})$, *then*

$$(x_n)_{n\in\mathbb{N}^+} \text{ is Cauchy} \Rightarrow \exists_{y\in\mathbb{R}}\big(x_n \xrightarrow{n} y\big)$$

$$\psi(x_n) \Rightarrow \exists_{y\in\mathbb{F}(\mathbb{N}^+,\mathbb{Q})}\exists_{E\in\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)}\exists_{H\in\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)}\big(E : y\in\mathbb{R}\ \&\ H : x_n \xrightarrow{n} y\big)\big)$$

Proof.

If $C : \mathrm{Cauchy}\big((x_n)_{n\in\mathbb{N}^+}\big)$, then $y : \mathbb{N}^+ \to \mathbb{Q}$, where

$$y_k := \big[x_{D(k)}\big]_{2k},$$

$$D(k) := 3k \vee C(2k),$$

is a Cauchy sequence, and $x_n \xrightarrow{n} y$. □

The above proof avoids the use of CC. Its standard use is due to the lack of witnessing-data in the formulation of the related concepts.

Primitives of BST I

- (s, t)
- equality $:=$ between terms
- $\mathtt{pr}_1(s, t) := s$ and $\mathtt{pr}_2(s, t) := t$
- $\mathbb{N}$.
- Any other totality X is defined through a "membership-formula" $x \in X$.
- A defined equality on X is a formula $x =_X y$ that satisfies the properties of an equivalence relation.
- If X is a set and Y is a totality, an assignment routine $\alpha : X \rightsquigarrow Y$ from X to Y is a finite routine assigning an element y of Y, to each given element x of X. In this case we write $\alpha(x) := y$.
- If X, Y are sets, an assignment routine $f : X \rightsquigarrow Y$ is a function, if $f(x) =_Y f(x')$, for every $x, x' \in X$, such that $x =_X x'$. In this case we write $f : X \to Y$.

Primitives of BST II

- $\mathbb{F}(X, Y)$ with pointwise equality is a set (definitional form of the function extensionality axiom).
- The universe of sets $\mathbb{V}_0$ with equality (definitional form of the univalence axiom)

$$X =_{\mathbb{V}_0} Y :\Leftrightarrow \exists_{f \in \mathbb{F}(X,Y)} \exists_{g \in \mathbb{F}(Y,X)} \big(g \circ f = \mathrm{id}_X \;\&\; f \circ g = \mathrm{id}_Y\big)$$

  is a class.
- If I is a set and $\mu_0 : I \rightsquigarrow \mathbb{V}_0$, a dependent assignment routine over μ_0 is an assignment routine μ_1 that assigns to each element i in I an element $\mu_1(i)$ in $\mu_0(i)$. We write such a routine by

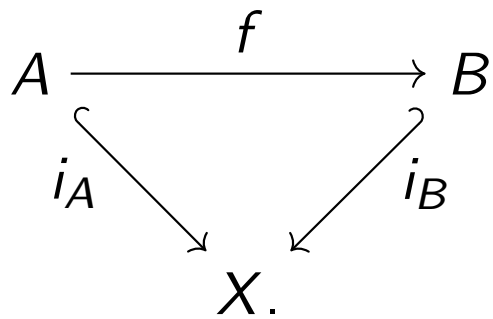
$$\mu_1 : \bigwedge_{i \in I} \mu_0(i),$$

  and their totality by $\mathbb{A}(I, \mu_0)$. If $\mu_1, \nu_1 : \bigwedge_{i \in I} \mu_0(i)$, we define

$$\mu_1 =_{\mathbb{A}(I,\mu_0)} \nu_1 :\Leftrightarrow \forall_{i \in I} \big(\mu_1(i) =_{\mu_0(i)} \nu_1(i)\big).$$

Primitives of BST III

- If X is a set, a subset of X is a pair (A, i_A), where A is a set and $i_A : A \hookrightarrow X$. If (A, i_A) and (B, i_B) are subsets of X, A is a subset of B, $A \subseteq B$, if there is $f : A \to B$ sttfdc

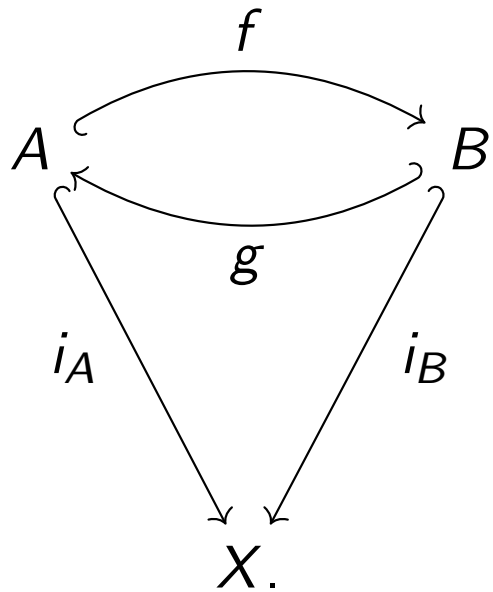


In this case we write $f : A \subseteq B$, and f is an embedding.

$$f : A \subseteq B \ \& \ h : A \subseteq B \Rightarrow f =_{\mathbb{F}(A,B)} h$$

$$(A, i_A) =_{\mathcal{P}(X)} (B, i_B) :\Leftrightarrow A \subseteq B \ \& \ B \subseteq A$$

$$(f, g) : A =_{\mathcal{P}(X)} B$$



$$(f, g) : A =_{\mathcal{P}(X)} B \Rightarrow (f, g) : A =_{\mathbb{V}_0} B.$$

$$(f, g) : A =_{\mathcal{P}(X)} B \ \& \ (f', g') : A =_{\mathcal{P}(X)} B \Rightarrow f = f' \ \& \ g = g'$$

Primitives of BST IV

If $P(x)$ is an extensional property on X i.e.,

$$\forall_{x,y\in X}\big(x =_X y \;\&\; P(x) \Rightarrow P(y)\big),$$

the set X_P generated by P is defined by the membership-condition

$$x \in X_P :\Leftrightarrow x \in X \;\&\; P(x),$$

and the canonical equality of X_P is inherited from the equality of X.

$$\{x \in X \mid P(x)\} := X_P$$

If X, Y are sets, their product $X \times Y$ is defined by

$$z \in X \times Y :\Leftrightarrow \exists_{x\in X}\exists_{y\in Y}\big(z := (x, y)\big),$$

and its equality is defined as usual.

If X is a set, then the property $P(x, y) :\Leftrightarrow x =_X y$ is an extensional property on $X \times X$ that generates the following subset of $X \times X$

$$D(X) := \big\{(x, y) \in X \times X \mid x =_X y\big\},$$

which we call the diagonal of X w.r.t. $=_X$.

Definitional form of the type-theoretic transport

If I is a set, a family of sets indexed by I is a pair $\Lambda := (\lambda_0, \lambda_1)$ s.t.

$$\lambda_0 : I \rightsquigarrow \mathbb{V}_0,$$

$$\lambda_1 : \bigwedge_{(i,j)\in D(I)} \mathbb{F}\big(\lambda_0(i), \lambda_0(j)\big),$$

(a) For every $i \in I$, we have that $\lambda_{ii} := \mathrm{id}_{\lambda_0(i)}$.
(b) If $i =_I j$ and $j =_I k$, the following diagram commutes

$$\begin{array}{ccc} \lambda_0(i) & & \\ {\scriptstyle \lambda_{ij}}\downarrow & \searrow^{\lambda_{ik}} & \\ \lambda_0(j) & \xrightarrow[\lambda_{jk}]{} & \lambda_0(k). \end{array}$$

If $i =_I j$, we call the function λ_{ij} the transport map from $\lambda_0(i)$ to $\lambda_0(j)$, and we call λ_1 the modulus of function-likeness of λ_0:

$$(\lambda_{ij}, \lambda_{ji}) : \lambda_0(i) =_{\mathbb{V}_0} \lambda_0(j).$$

The exterior union of Λ, or the $\sum$-set of Λ

$$w \in \sum_{i \in I} \lambda_0(i) :\Leftrightarrow \exists_{i \in I} \exists_{x \in \lambda_0(i)} \big(w := (i, x)\big),$$

$$(i, x) =_{\sum_{i \in I} \lambda_0(i)} (j, y) :\Leftrightarrow i =_I j \;\&\; \lambda_{ij}(x) =_{\lambda_0(j)} y.$$

We get by definition the proof-irrelevant version of the path-lifting-property (Lemma 2.3.2. and Theorem 2.7.2 in book-HoTT).

The assignment routine $\mathrm{pr}_1^{\Lambda} : \sum_{i \in I} \lambda_0(i) \rightsquigarrow I$,

$$\mathrm{pr}_1^{\Lambda}(i, x) := \mathrm{pr}_1(i, x) := i$$

is a function.

The dependent functions over Λ, or the $\prod$-set of Λ

$$\Phi \in \prod_{i \in I} \lambda_0(i) :\Leftrightarrow \Phi : \bigwedge_{i \in I} \lambda_0(i) \;\; \& \;\; \forall_{(i,j) \in D(I)} \big(\Phi_j =_{\lambda_0(j)} \lambda_{ij}(\Phi_i)\big),$$

and it is equipped with the pointwise equality of $\mathbb{A}(I, \lambda_0)$.

Definitional and proof-irrelevant formulation of dependent application (Lemma 2.3.4. in book-HoTT, proof with J)

$$\mathtt{apd}_\Phi : \prod_{p : i =_I j} \big(p_*(\Phi_i) =_{\lambda_0(j)} \Phi_j\big),$$

$$p_* := \lambda_{ij}.$$

The second projection of the $\sum$-set of Λ

The pair $\Sigma^{\Lambda} := (\sigma_0^{\Lambda}, \sigma_1^{\Lambda})$ is a family over $\sum_{i \in I} \lambda_0(i)$, where

$$\sigma_0^{\Lambda} : \sum_{i \in I} \lambda_0(i) \rightsquigarrow \mathbb{V}_0,$$

$$\sigma_0^{\Lambda}(i, x) := \lambda_0(i),$$

$$\sigma_1^{\Lambda}\big((i, x), (j, y)\big) := \lambda_{ij}.$$

The dependent assignment routine

$$\mathtt{pr}_2^{\Lambda} : \bigwedge_{(i,x) \in \sum_{i \in I} \lambda_0(i)} \lambda_0(i),$$

$$\mathtt{pr}_2^{\Lambda}(i, x) := \mathtt{pr}_2(i, x) := x,$$

is a dependent function over Σ^{Λ}: if $(i, x) =_{\sum_{i \in I} \lambda_0(i)} (j, y)$, then

$$\mathtt{pr}_2^{\Lambda}(j, y) := y =_{\lambda_0(j)} \lambda_{ij}(x) := \lambda_{ij}\big(\mathtt{pr}_2^{\Lambda}(i, x)\big).$$

Distributivity of $\prod$ over $\sum$ (In lccc, Martin-Löf, Awodey)

Let X, Y be sets,

(ρ_0, ρ_1) is an $(X \times Y)$-family of sets,

If $x \in X$, let $(\lambda_0^x, \lambda_1^x)$ is the Y-family

$$\lambda_0^x(y) := \rho_0(x, y),\ \lambda_1^x : \rho_0(x, y) \to \rho_0(x, y'),\ \lambda_{yy'}^x := \rho_{(x,y)(x,y')}$$

Let the X-family of sets (μ_0, μ_1), where

$$\mu_0(x) := \sum_{y \in Y} \rho_0(x, y),$$

$$\mu_1 : \bigwedge_{(x,x') \in D(X)} \mathbb{F}\left(\sum_{y \in Y} \rho_0(x, y), \sum_{y \in Y} \rho_0(x', y) \right)$$

$$\mu_{xx'}(y, u) := \left(y, \rho_{(x,y)(x',y)}(u)\right).$$

Lemma

If $\Phi \in \prod_{x\in X} \mu_0(x) := \prod_{x\in X} \sum_{y\in Y} \rho_0(x,y)$, *the a.r.* $f_\Phi : X \rightsquigarrow Y$, $x \mapsto \mathtt{pr}_1(\Phi_x)$, *is a function from* X *to* Y.

Lemma

If $f : X \to Y$ *the pair* $\left(\nu_0^f, \nu_1^f\right)$ *is an* X*-family of sets, where*

$$\nu_0(x) := \rho_0(x, f(x)),$$

$$\nu_1^f : \bigwedge_{(x,x')\in D(X)} \mathbb{F}\left(\rho_0(x,f(x)), \rho_0(x',f(x'))\right), \quad \nu^f_{xx'} := \rho_{(x,f(x))(x',f(x'))}.$$

Lemma

The pair (ξ_0, ξ_1) *is an* $\mathbb{F}(X,Y)$*-family of sets, where*

$$\xi_0(f) := \prod_{x\in X} \nu_0^f(x) := \prod_{x\in X} \rho_0(x, f(x)),$$

$$\xi_{ff'} : \prod_{x\in X} \rho_0(x,f(x)) \to \prod_{x\in X} \rho_0(x,f'(x)), [\xi_{ff'}(H)]_x := \rho_{(x,f(x))(x,f'(x))}(H_x)$$

Theorem

If

$$\Phi \in \prod_{x \in X} \mu_0(x) := \prod_{x \in X} \sum_{y \in Y} \rho_0(x, y),$$

there is

$$\Theta_\Phi \in \prod_{x \in X} \nu_0^{f_\Phi} := \prod_{x \in X} \rho_0(x, f_\Phi(x))$$

such that

$$\left(f_\Phi, \Theta_\Phi\right) \in \sum_{f \in \mathbb{F}(X,Y)} \prod_{x \in X} \rho_0(x, f_\Phi(x))$$

and the following a.r. is a function:

$$\mathtt{ac} : \prod_{x \in X} \sum_{y \in Y} \rho_0(x, y) \rightsquigarrow \sum_{f \in \mathbb{F}(X,Y)} \prod_{x \in X} \rho_0(x, f(x)),$$

$$\Phi \mapsto \left(f_\Phi, \Theta_\Phi\right).$$

From MLTT + FunExt-proofs to BST-proofs using the canonical equality on the function space

The simplest example: if $X : \mathcal{U}$, then by FunExt

$$\mathrm{id}_X =_{X \to X} \mathrm{id}_X \leftrightarrow \prod_{x:X}(x =_X x)$$

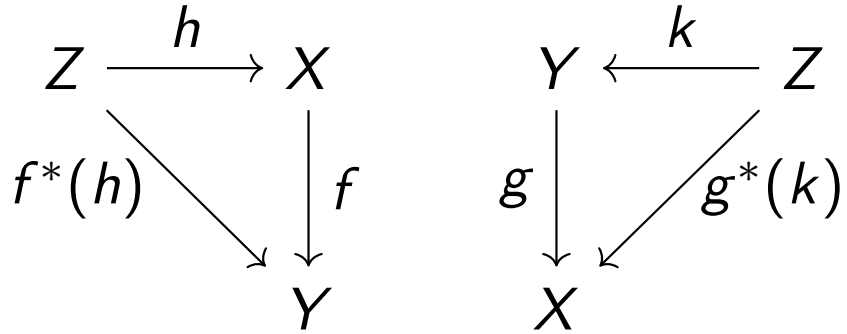
In BISH we have by definition that

$$\mathrm{id}_X =_{\mathbb{F}(X,X)} \mathrm{id}_X :\Leftrightarrow \forall_{x \in X}(x =_X x)$$

From MLTT + UA-proofs to BST-proofs using the canonical equality on the universe of sets

Proposition (lemma 4.9.2. in book HoTT)

If $(f, g) : X =_{\mathbb{V}_0} Y$, *then* $(f^*, g^*) : \mathbb{F}(Z, X) =_{\mathbb{V}_0} \mathbb{F}(Z, Y)$



Proof of Lemma 4.9.2: If $e : X \simeq Y$, then $Z \to X \simeq Z \to Y$.

By UA we get $e = \mathtt{idtoeqv}(p)$, for some $p : X =_{\mathcal{U}} Y$.

In the formulation of this lemma the universe-level is not raised.

Some proof-irrelevant definitions for a set $(X, =_X)$

X is a inhabited, if $\exists_{x \in X}(x =_X x)$.

X is a singleton, or contractible, or a (-2)-set, if

$$\exists_{x_0 \in X} \forall_{x \in X}(x_0 =_X x).$$

X is contractible iff $X =_{\mathbb{V}_0} \mathbf{1}$.

X is a subsingleton, or a mere proposition, or a (-1)-set, if

$$\forall_{x,y \in X}(x =_X y).$$

The truncation of X is the set $(X, {}_{||=_X||})$,

$$x \;{}_{||=_X||}\; y :\Leftrightarrow x =_X x \;\&\; y =_X y.$$

The definition of a 0-set requires to determine a set

$$\mathtt{PrfEq}^X(x, y)$$

of witnesses of the equality $x =_X y$. This is done in a universal way in MLTT, while in BST in a "local" way, and by definition.

$$(f, g) : X =_{\mathbb{V}_0} Y \Leftrightarrow \forall_{y \in Y} \big(\mathtt{fib}^f(y) \text{ is contractible} \big)$$

$$\mathtt{fib}^f(y) := \{x \in X \mid f(x) =_Y y\}$$

$(\Rightarrow)$ $g(y)$ is the center of contraction

$(\Leftarrow)$ $g : Y \rightsquigarrow X,\ y \mapsto x_y,$ x_y is the center of contraction of $\mathtt{fib}^f(y)$

$$\forall_{x \in \mathtt{fib}^f(y)} \big(x =_X x_y \big)$$

g is a function

$$f(g(y) := f(x_y) =_Y y$$

$$g(f(x)) := x_{f(x)} =_X x,$$

since $x \in \mathtt{fib}^f(f(x))$.

Corollary 4.9.3. and Theorem 4.9.4. in book-HoTT

Proposition

$\Lambda := (\lambda_0, \lambda_1)$ *is an* I*-family of sets s.t.* $\lambda_0(i)$ *is contractible,* $i \in I$*.*

$$(i) \qquad (\mathtt{pr}_1^{\Lambda}, \phi) : \sum_{i \in I} \lambda_0(i) =_{\mathbb{V}_0} I$$

$$\phi(i) := (i, x_i), \quad x_i \text{ is the center of contraction for } \lambda_0(i).$$

$$(ii) \qquad (\Phi, \Theta) : \mathbb{F}\left(I, \sum_{i \in I} \lambda_0(i)\right) =_{\mathbb{V}_0} \mathbb{F}(I, I).$$

$$(iii) \qquad \prod_{i \in I} \lambda_0(i) \text{ is a retract of } \mathtt{fib}^{\Phi}(\mathrm{id}_I)$$

Sketch of the proof, case (ii)

$$\Phi : \mathbb{F}\left(I, \sum_{i\in I} \lambda_0(i)\right) \rightsquigarrow \mathbb{F}(I, I)$$

$$f \mapsto \Phi(f)$$

$$I \xrightarrow{f} \textstyle\sum_{i\in I} \lambda_0(i) \xrightarrow{\mathtt{pr}_1^{\Lambda}} I$$

$$\Phi(f)$$

$$\Theta : \mathbb{F}(I, I) \rightsquigarrow \mathbb{F}\left(I, \sum_{i\in I} \lambda_0(i)\right)$$

$$g \mapsto \Theta(g)$$

$$[\Theta(G)](i) := \left(g(i), x_{g(i)}\right)$$

Multiple use of the canonical equality on function space (FunExt).

Sketch of the proof, case (iii)

$$\mathtt{fib}^{\Phi}(\mathrm{id}_I) := \left\{ f \in \mathbb{F}\left(I, \sum_{i \in I} \lambda_0(i)\right) \mid \Phi(f) =_{\mathbb{F}(I,I)} \mathrm{id}_I \right\}$$

$$\prod_{i \in I} \lambda_0(i) \xrightarrow{s^{\Phi}} \mathtt{fib}^{\Phi}(\mathrm{id}_I) \xrightarrow{r^{\Phi}} \prod_{i \in I} \lambda_0(i)$$

$$\mathrm{id}_{\prod_{i \in I} \lambda_0(i)}$$

$$[r^{\Phi}(f)]_i := \lambda_{\mathtt{pr}_1^{\wedge}(f(i))i}\left(\mathtt{pr}_2^{\wedge}(f(i))\right)$$

$$[\Phi(f)](i) := \mathtt{pr}_1^{\wedge}\left(f(i)\right) =_I \mathrm{id}_I(i) := i$$

$$[s^{\Phi}(\Psi)](i) := (i, \Psi_i)$$

Multiple use of the canonical equality on function space (FunExt).

Bishop's definition of the set $\mathbb{R}$ of reals

$$\mathbb{R} := \left\{ x \in \mathbb{F}(\mathbb{N}^+, \mathbb{Q}) \mid \forall_{m,n \in \mathbb{N}^+} \left(|x_m - x_n| \leq \frac{1}{m} + \frac{1}{n} \right) \right\}$$

$$x =_{\mathbb{R}} y :\Leftrightarrow \forall_{n \in \mathbb{N}^+} \left(|x_n - y_n| \leq \frac{2}{n} \right)$$

$$x =_{\mathbb{R}} y \Leftrightarrow \forall_{j \in \mathbb{N}^+} \exists_{N_j \in \mathbb{N}^+} \forall_{n \geq N_j} \left(|x_n - y_n| \leq \frac{1}{j} \right)$$

$$x =_{\mathbb{R}} y \overset{(\mathrm{CC})}{\Leftrightarrow} \exists_{\omega \in \mathbb{F}(\mathbb{N}^+, \mathbb{N}^+)} \forall_{j \in \mathbb{N}^+} \forall_{n \geq \omega(j)} \left(|x_n - y_n| \leq \frac{1}{j} \right)$$

A proof-relevant membership-condition for $\mathbb{R}$

$$x \in \mathbb{R} :\Leftrightarrow x \in \mathbb{F}(\mathbb{N}^+, \mathbb{Q}) \;\;\&\;\; \exists_{C \in \mathbb{F}(\mathbb{N}^+, \mathbb{N}^+)} \big(C : x \in \mathbb{R}\big)$$

$$C : x \in \mathbb{R} :\Leftrightarrow \forall_{k \in \mathbb{N}^+} \forall_{m,n \geq C(k)} \left(|x_m - x_n| \leq \frac{1}{k} \right)$$

If $x \in \mathbb{F}(\mathbb{N}^+, \mathbb{Q})$, we define

$$\mathtt{PrfMemb}_0^{\mathbb{R}}(x) := \big\{ C \in \mathbb{F}(\mathbb{N}^+, \mathbb{N}^+) \mid C : x \in \mathbb{R} \big\}$$

A proof-relevant equality $=_{\mathbb{R}}$

$$\mathtt{PrfEq}^{\mathbb{R}}(x,y) := \left\{\omega \in \mathbb{F}(\mathbb{N}^+,\mathbb{N}^+) \mid \omega : x =_{\mathbb{R}} y\right\}$$

$$\omega : x =_{\mathbb{R}} y :\Leftrightarrow \omega \geq \mathrm{id}_{\mathbb{N}^+} \;\&\; \forall_{j\in\mathbb{N}^+}\forall_{n\geq\omega(j)}\left(|x_n - y_n| \leq \frac{1}{j}\right)$$

$$x =_{\mathbb{R}} y :\Leftrightarrow \exists_{\omega\in\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)}\left(\omega : x =_{\mathbb{R}} y\right)$$

$\mathtt{PrfEq}^{\mathbb{R}}(x,y)$, equipped with its canonical equality, is not a (-1)-set. It becomes, if we truncate $\mathtt{PrfEq}^{\mathbb{R}}(x,y)$

$$\omega \;||=_{\mathbb{R}}||\; \delta :\Leftrightarrow \omega =_{\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)} \omega \;\;\&\;\; \delta =_{\mathbb{F}(\mathbb{N}^+,\mathbb{N}^+)} \delta$$

If $\omega \in \mathtt{PrfEq}^{\mathbb{R}}(x,y)$ and $\delta \in \mathtt{PrfEq}^{\mathbb{R}}(y,z)$, we define

$$\mathtt{refl}(x) := \mathrm{id}_{\mathbb{N}^+} \;\;\&\;\; \omega^{-1} := \omega \;\;\&\;\; (\omega * \delta)(j) := \omega(2j) \vee \delta(2j)$$

Extensional subsets equipped with an $\exists$-equality condition

X is a set

$Y := X_P$

Z is a set

Y is equipped with the equality (not the one inherited from X)

$$x =_Y x' :\Leftrightarrow \exists_{p \in Z}\big(p : x =_Y x'\big)$$

$$\theta^{xx'}(p) :\Leftrightarrow p : x =_Y x' \text{ is an extensional property on } Z$$

$$\mathtt{PrfEq}^Y(x, x') := \{p \in Z \mid p : x =_Y x'\}$$

$$\mathtt{refl}(x) \in \mathtt{PrfEq}^Y(x, x)$$

$$\mathtt{PrfEq}^Y(x, x') \ni p \mapsto p^{-1} \in \mathtt{PrfEq}^Y(x', x)$$

$$\Big(p \in \mathtt{PrfEq}^Y(x, x'), q \in \mathtt{PrfEq}^Y(x', x'')\Big) \mapsto p{*}q \in \mathtt{PrfEq}^Y(x, x'')$$

Equality of sets

$$X =_{\mathbb{V}_0} Y :\Leftrightarrow \exists_{f\in\mathbb{F}(X,Y)}\exists_{g\in\mathbb{F}(Y,X)}\big(g\circ f = \mathrm{id}_X \;\&\; f\circ g = \mathrm{id}_Y\big)$$

$$\mathtt{PrfEq}(X,Y) := \big\{(f,g)\in\mathbb{F}(X,Y)\times\mathbb{F}(Y,X)\mid (f,g) : X =_{\mathbb{V}_0} Y\big\}$$

$$(\mathrm{id}_2,\mathrm{id}_2)\in\mathtt{PrfEq}(2,2)\quad\&\quad(\mathtt{sw}_2,\mathtt{sw}_2)\in\mathtt{PrfEq}(2,2)$$

$$\mathtt{refl}(X) := \big(\mathrm{id}_X,\mathrm{id}_X\big)$$

$$(f,g)^{-1} := (g,f)$$

$$(f,g)*(h,k) := (h\circ f, g\circ k)$$

Equality of subsets of a set X

$$\mathtt{PrfEq}(A,B) := \left\{(f,g) \in \mathbb{F}(A,B)\times\mathbb{F}(B,A) \mid f : A \subseteq B \;\&\; g : B \subseteq A\right\}$$

$$(f,g) : A =_{\mathcal{P}(X)} B \;\&\; (f',g') : A =_{\mathcal{P}(X)} B \Rightarrow (f,g) =_{\mathtt{PrfEq}(A,B)} (f',g')$$

$$\mathtt{refl}(A) := \left(\mathrm{id}_A, \mathrm{id}_A\right)$$

$$(f,g)^{-1} := (g,f)$$

$$(f,g) * (h,k) := (h \circ f, g \circ k),$$

The (impredicative) Richman ordinals

If $<$ is a binary relation on a set W, $H \subseteq W$ is hereditary, if

$$\forall_{w \in W}\Big(\forall_{u \in W}\big(u < w \Rightarrow u \in H\big) \Rightarrow w \in H\Big).$$

The relation $<$ is well-founded, if

$$\forall_{H \in \mathcal{P}(X)}\big(H \text{ is hereditary } \Rightarrow H = W\big).$$

A Richman ordinal is a pair $(\alpha, \leq)$, where α is discrete ($x =_\alpha y \vee x \neq_\alpha y$, and $x \neq_\alpha y$ is a given apartness relation on X) $\leq$ is a linear order ($x \leq y \vee y \leq x$, for every $x, y \in \alpha$), and $<$ is well founded, where $x < y :\Leftrightarrow x \leq y \;\&\; x \neq_\alpha y$. An injection $\rho : \alpha \leq \beta$ from α to β is a function $\rho : \alpha \to \beta$ such that

(i) $\forall_{x,y \in \alpha}\big(x < y \Rightarrow \rho(x) < \rho(y)\big)$.

(ii) $\forall_{z \in \beta}\forall_{y \in \alpha}\big(z < \rho(y) \Rightarrow \exists_{x \in \alpha}(\rho(x) =_\beta z)\big)$.

[9], p. 28: There is at most one injection from α to β.

Equality of Richman ordinals

$$\alpha =_{\mathrm{Ord}_R} \beta :\Leftrightarrow \alpha \leq \beta \;\&\; \beta \leq \alpha$$

$$\mathtt{PrfEq}(\alpha,\beta) := \{(\rho,\sigma) \in \mathbb{F}(\alpha,\beta)\times\mathbb{F}(\beta,\alpha) \mid \rho : \alpha \leq \beta \;\&\; \sigma : \beta \leq \alpha\}$$

$$(\rho,\sigma) : \alpha =_{\mathrm{Ord}_R} \beta \;\&\; (\rho',\sigma') : \alpha =_{\mathrm{Ord}_R} \beta \Rightarrow \rho =_{\mathbb{F}(\alpha,\beta)} \rho' \;\&\; \sigma =_{\mathbb{F}(\beta,\alpha)} \sigma'$$

$$\mathtt{refl}(\alpha) := \left(\mathrm{id}_\alpha, \mathrm{id}_\alpha\right)$$

$$(\rho,\sigma)^{-1} := (\sigma,\rho)$$

$$(\rho,\sigma) * (\tau,\nu) := (\tau \circ \rho, \sigma \circ \nu)$$

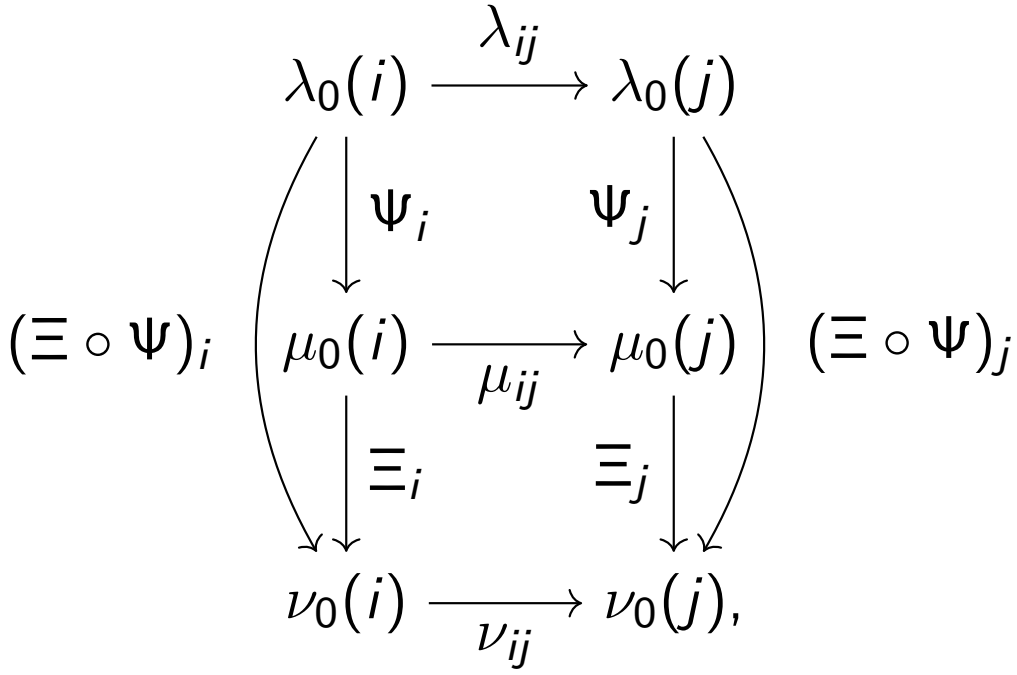
Family-maps

Let $\Lambda := (\lambda_0, \lambda_1)$ and $M := (\mu_0, \mu_1)$ be I-families of sets. A family-map from Λ to M is a d.a.r.

$$\Psi : \bigsqcap_{i \in I} \mathbb{F}\big(\lambda_0(i), \mu_0(i)\big)$$

$$\begin{array}{ccc} \lambda_0(i) & \xrightarrow{\lambda_{ij}} & \lambda_0(j) \\ \Big\downarrow{\scriptstyle \Psi_i} & & \Big\downarrow{\scriptstyle \Psi_j} \\ \mu_0(i) & \xrightarrow[\mu_{ij}]{} & \mu_0(j). \end{array}$$

$$\Psi =_{\mathtt{Map}_I(\Lambda, M)} \Xi :\Leftrightarrow \forall_{i \in I} \big(\Psi_i =_{\mathbb{F}(\lambda_0(i), \mu_0(i))} \Xi_i\big).$$



$$\mathrm{Id}_\Lambda : \bigwedge_{i\in I} \mathbb{F}\big(\lambda_0(i), \lambda_0(i)\big), \quad \mathrm{Id}_\Lambda(i) := \mathrm{id}_{\lambda_0(i)}, \quad i \in I$$

Equality of I-families

$$\Lambda =_{\mathtt{Fam}(I)} M :\Leftrightarrow \exists_{\Phi\in\mathtt{Map}_I(\Lambda,M)}\exists_{\Psi\in\mathtt{Map}_I(M,\Lambda)}\big(\Phi\circ\Psi = \mathrm{id}_M \;\&\; \Psi\circ\Phi = \mathrm{id}_\Lambda\big)$$

$$\mathtt{PrfEq}(\Lambda, M) := \big\{(\Phi,\Psi) \in \mathtt{Map}_I(\Lambda, M) \times \mathtt{Map}_I(M,\Lambda) \mid$$

$$(\Phi,\Psi) : \Lambda =_{\mathtt{Fam}(I)} M\big\}$$

In general, not all elements of $\mathtt{PrfEq}(\Lambda, M)$ are equal.

$$\mathtt{refl}(\Lambda) := \big(\mathrm{Id}_\Lambda, \mathrm{Id}_\Lambda\big)$$

$$(\Phi,\Psi)^{-1} := (\Psi,\Phi)$$

$$(\Phi,\Psi) * (\Phi',\Psi') := (\Phi'\circ\Phi, \Psi\circ\Psi')$$

The equality of I-families of subsets behaves like the equality of subsets.

Equality of integrable functions in BCMT

$\mathfrak{F}(X)$ is the totality of real-valued, str. ext., part. functions on X
$\mathcal{L} := (X, L, \int)$ is an integration space

$$L_1 := \{f \in \mathfrak{F}(X) \mid f \text{ has a representation in } L\}$$

$(f_n)_{n=1}^{\infty}$ is a sequence of partial functions in L such that

$$\sum_{n\in\mathbb{N}^+} \int |f_n| < +\infty, \quad \text{and}$$

$$\forall_{x\in X}\left(\sum_{n\in\mathbb{N}^+} |f_n(x)| < +\infty \Rightarrow f(x) = \sum_{n\in\mathbb{N}^+} f_n(x) \right)$$

$F \subseteq X$ is full, if there is $g \in L_1$ s.t. $\mathrm{dom} g \subseteq F$.

$$f =_{L_1} g :\Leftrightarrow \exists_{F\in\mathcal{P}(X)}\big(F \text{ is full } \&\ f_{|F} = g_{|F}\big)$$

$$\mathtt{refl}(f) := \mathrm{dom}(f) \quad \& \quad F^{-1} := F \quad \& \quad F * G := F \cap G$$

A set X with a proof-relevant equality

$$x =_X x' :\Leftrightarrow \exists_{p \in Z}\left(p : x =_X x'\right)$$

$$\theta^{xx'}(p) :\Leftrightarrow p : x =_Y x' \text{ is an extensional property on } Z$$

$$\mathtt{PrfEq}^X(x, x') := \{p \in Z \mid p : x =_X x'\}$$

$$\mathtt{refl} : \bigwedge_{x \in X} \mathtt{PrfEq}^X(x, x)$$

$$^{-1} : \bigwedge_{x,x' \in D(X)} \bigwedge_{p \in \mathtt{PrfEq}^X(x,x')} \mathtt{PrfEq}^X(x', x)$$

$$* : \bigwedge_{x,x',x'' \in D(X)} \bigwedge_{p \in \mathtt{PrfEq}^X(x,x')} \bigwedge_{q \in \mathtt{PrfEq}^X(x',x'')} \mathtt{PrfEq}^X(x, x'')$$

A set X is a Martin-Löf set, if

(i) $\mathtt{refl}(x) * p =_{\mathtt{PrfEq}^X(x,x')} p$ and $p * \mathtt{refl}(y) =_{\mathtt{PrfEq}^X(x,x')} p$,
for every $p \in \mathtt{PrfEq}^X(x, x')$.

(ii) $p * p^{-1} =_{\mathtt{PrfEq}^X(x,x)} \mathtt{refl}(x)$ and $p^{-1} * p =_{\mathtt{PrfEq}^X(y,y)} \mathtt{refl}(y)$,
for every $p \in \mathtt{PrfEq}^X(x, x')$.

(iii) $(p * q) * r =_{\mathtt{PrfEq}(x,x''')} p * (q * r)$,
for every $p \in \mathtt{PrfEq}^X(x, x'), q \in \mathtt{PrfEq}^X(x', x'')$ and
$r \in \mathtt{PrfEq}^X(x'', x''')$.

(iv) If $p, q \in \mathtt{PrfEq}^X(x, x')$ and $r, s \in \mathtt{PrfEq}^X(x', x'')$, then

$$p =_{\mathtt{PrfEq}^X(x,x')} q \;\&\; r =_{\mathtt{PrfEq}^X(x',x'')} s \Rightarrow p * r =_{\mathtt{PrfEq}^X(x,x'')} q * s$$

A set X with a proof-relevant equality is a 0-set, if

$$\forall_{x,y \in X} \forall_{p,q \in \mathtt{PrfEq}^X(x,y)} \left(p =_{\mathtt{PrfEq}^X(x,y)} q\right)$$

Proof-relevant family of sets $\Lambda := (\lambda_0, \lambda_1)$ over I

$$\lambda_0 : I \rightsquigarrow \mathbb{V}_0, \qquad \lambda_1 : \bigwedge_{i,j \in I} \bigwedge_{p \in \mathtt{PrfEq}^I(i,j)} \mathbb{F}\big(\lambda_0(i)\lambda_0(j)\big)$$

$$\lambda_1\big(i,j,p\big) := \lambda_{ij}^{p}$$

$$\lambda_{ii}^{\mathtt{refl}(i)} = \mathrm{id}_{\lambda_0(i)}$$

$$\begin{array}{ccc} \lambda_0(i) & & \\ {\scriptstyle \lambda_{ij}^{p}}\downarrow & \searrow^{\lambda_{ik}^{p*q}} & \\ \lambda_0(j) & \xrightarrow[\lambda_{jk}^{q}]{} & \lambda_0(k). \end{array}$$

$$\begin{array}{ccc} \lambda_0(i) & & \\ {\scriptstyle \lambda_{ij}^{p}}\downarrow & \searrow^{\mathrm{id}_{\lambda_0(i)}} & \\ \lambda_0(j) & \xrightarrow[\lambda_{ji}^{p^{-1}}]{} & \lambda_0(i) \end{array} \qquad \begin{array}{ccc} \lambda_0(j) & & \\ {\scriptstyle \lambda_{ji}^{p^{-1}}}\downarrow & \searrow^{\mathrm{id}_{\lambda_0(j)}} & \\ \lambda_0(i) & \xrightarrow[\lambda_{ij}^{p}]{} & \lambda_0(i). \end{array}$$

Λ is function-like, if for every $i, j \in I$ and every $p, q \in \mathtt{PrfEq}(i, j)$

$$p =_{\mathtt{PrfEq}'(i,j)} p' \Rightarrow \lambda_{ij}^{p} =_{\mathbb{F}(\lambda_0(i),\lambda_0(j))} \lambda_{ij}^{p'}$$

Λ is proof-irrelevant, if for every $i, j \in I$ and $p, p' \in \mathtt{PrfEq}'(i, j)$

$$\lambda_{ij}^{p} =_{\mathbb{F}(\lambda_0(i),\lambda_0(j))} \lambda_{ij}^{p'}$$

The $\sum_{i \in \boldsymbol{I}} \lambda_0(i)$-set of a proof-relevant family of sets Λ

$$(i, x) =_{\sum_{i \in \boldsymbol{I}} \lambda_0(i)} (j, y) :\Leftrightarrow i =_{\boldsymbol{I}} j \;\&\; \exists_{p \in \mathtt{PrfEq}^{\boldsymbol{I}}(i,j)} \left(\lambda^p_{ij}(x) =_{\lambda_0(j)} y\right)$$

If Λ is function-like, $\sum_{i \in \boldsymbol{I}} \lambda_0(i)$ has a proof-relevant equality

$$Q^{xy}_{ij}(p) :\Leftrightarrow \lambda^p_{ij}(x) = y \;\text{ is extensional on } \mathtt{PrfEq}^{\boldsymbol{I}}(i, j)$$

$$\mathtt{PrfEq}^{\sum_{i \in \boldsymbol{I}} \lambda_0(i)}\big((i, x), (j, y)\big) := \left\{p \in \mathtt{PrfEq}^{\boldsymbol{I}}(i, j) \mid \lambda^p_{ij}(x) = y\right\}$$

$$\mathtt{refl}(i, x) := \mathtt{refl}(i)$$

If $\boldsymbol{I}$ is a Martin-Löf set, then $\sum_{i \in \boldsymbol{I}} \lambda_0(i)$ is Martin-Löf.

Lemma

Let $\boldsymbol{X}$ *be a Martin-Löf set,* $x_0 \in X$ *and* $\mathtt{PrfEq}_0^X : X \rightsquigarrow \mathbb{V}_0$,

$$x \mapsto \mathtt{PrfEq}^X(x, x_0)$$

$$\mathtt{PrfEq}_1^X : \bigwedge_{x,y \in X} \bigwedge_{p \in \mathtt{PrfEq}^X(x,y)} \mathbb{F}\left(\mathtt{PrfEq}^X(x, x_0), \mathtt{PrfEq}^X(y, x_0)\right)$$

$$\mathtt{PrfEq}_1^X\left(x, y, p\right) := \mathtt{PrfEq}_{xy}^p : \mathtt{PrfEq}^X(x, x_0) \to \mathtt{PrfEq}^X(y, x_0)$$

$$r \mapsto p^{-1} * r$$

The pair $E := (\mathtt{PrfEq}_0^X, \mathtt{PrfEq}_1^X)$ *is a function-like, proof-relevant family of sets over* $\boldsymbol{X}$.

The "contractibility of singleton types"

Theorem

Let $\boldsymbol{X}$ *be a Martin-Löf set,* $x_0 \in X$ *and let* $E := (\mathtt{PrfEq}_0^X, \mathtt{PrfEq}_1^X)$ *be the previous function-like, proof-relevant family of sets over* $\boldsymbol{X}$*. Let* $\sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^X(x, x_0)$ *be the corresponding Martin-Löf set. Then for every* $(x, p) \in \sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^X(x, x_0)$ *we have that*

$$(x, p) =_{\sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^X(x, x_0)} \big(x_0, \mathtt{refl}(x_0)\big).$$

Proof.

$$(x, p) =_{\sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^X(x, x_0)} \big(x_0, \mathtt{refl}(x_0)\big) :\Leftrightarrow$$

$$:\Leftrightarrow x =_X x_0 \;\&\; \exists_{q \in \mathtt{PrfEq}^X(x, x_0)} \big(\mathtt{PrfEq}^q_{xx_0}(p) = \mathtt{refl}(x_0)\big).$$

If $(x, p) \in \sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^X(x, x_0)$, then $p \in \mathtt{PrfEq}^X(x, x_0)$, hence $x =_X x_0$. If $q := p$, then $\mathtt{PrfEq}^p_{xx_0}(p) := p^{-1} * p = \mathtt{refl}(x_0)$. □

$\boldsymbol{X}$, $\boldsymbol{Y}$ sets with proof-relevant equalities. A proof-relevant function from $\boldsymbol{X}$ to $\boldsymbol{Y}$ is a pair $\boldsymbol{f} := (f, f_1)$, where $f : X \to Y$ and

$$f_1 : \bigwedge_{x,x' \in X} \bigwedge_{p \in \mathtt{PrfEq}^X(x,x')} \mathtt{PrfEq}^Y(f(x), f(x'))$$

$\boldsymbol{f}$ is a Martin-Löf function, if

(i) $f_1(x, x, \mathtt{refl}(x)) =_{\mathtt{PrfEq}^Y(f(x),f(x))} \mathtt{refl}(f(x))$, for every $x \in X$.
(ii) If $x =_X x' =_X x''$, then
$f_1(x, x'', p * q) =_{\mathtt{PrfEq}^Y(f(x),f(x''))} f_1(x, x', p) * f_1(x', x'', q)$.

f_1 is the first associate of $\boldsymbol{f}$. If

$$p =_{\mathtt{PrfEq}^X(x,x')} p' \Rightarrow f_1(x, x', p) =_{\mathtt{PrfEq}^Y(f(x),f(x'))} f_1(x, x', p'),$$

f_1 is a function-like first associate of $\boldsymbol{f}$.

The pair $\mathtt{pr}_1^{\wedge} := (\mathtt{pr}_1^{\wedge}, \varpi_1)$ is a Martin-Löf function

$$\mathtt{pr}_1^{\wedge} : \sum_{i \in I} \lambda_0(i) \to I$$

$$\varpi_1 : \bigwedge_{(i,x),(j,y) \in \sum_{i \in I} \lambda_0(i)} \; \bigwedge_{p \in \mathtt{PrfEq}^{\sum_{i \in I} \lambda_0(i)}((i,x),(j,y))} \mathtt{PrfEq}^{I}(i,j)$$

$$\varpi_1((i,x),(j,y),p) := p$$

with a function-like first associate ϖ_1.

Let $\boldsymbol{X}, \boldsymbol{Y}$ be Martin-Löf sets, and $\boldsymbol{f} := (f, f_1) : \boldsymbol{X} \to \boldsymbol{Y}$ a Martin-Löf function with a function-like first associate f_1.

(i) If $y \in Y$, the pair $\mathtt{PrfEq}f := \left(\mathtt{PrfEq}f_0^y, \mathtt{PrfEq}f_1^y\right)$, where

$$x \mapsto \mathtt{PrfEq}^Y(f(x), y),$$

$$\mathtt{PrfEq}f_1^y : \bigwedge_{x,x' \in X} \bigwedge_{p \in \mathtt{PrfEq}(x,x')} \mathbb{F}\left(\mathtt{PrfEq}^Y(f(x), y), \mathtt{PrfEq}^Y(f(x'), y)\right)$$

$$\mathtt{PrfEq}f_1^y(x, x', p) := \mathtt{PrfEq}f_{xx'}^{y,p} : \mathtt{PrfEq}^Y(f(x), y) \to \mathtt{PrfEq}^Y(f(x'), y)$$

$$r \mapsto [f_1(x, x', p)]^{-1} * r,$$

for every $r \in \mathtt{PrfEq}^Y(f(x), y)$, is a function-like, proof-relevant family of sets over $\boldsymbol{X}$.

The proof-relevant fiber of a Martin-Löf function

(ii) The pair $\mathtt{fib}^{\boldsymbol{f}} := (\mathtt{fib}_0^{\boldsymbol{f}}, \mathtt{fib}_1^{\boldsymbol{f}})$, where $\mathtt{fib}_0^{\boldsymbol{f}} : \boldsymbol{Y} \rightsquigarrow \mathbb{V}_0$ is the rule

$$y \mapsto \mathtt{fib}^{\boldsymbol{f}}(y) := \sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^{\boldsymbol{Y}}(f(x), y),$$

$$\mathtt{fib}_1^{\boldsymbol{f}} : \bigwedge_{y,y' \in \boldsymbol{Y}} \bigwedge_{q \in \mathtt{PrfEq}^{\boldsymbol{Y}}(y,y')} \mathbb{F}\left(\sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^{\boldsymbol{Y}}(f(x), y), \sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^{\boldsymbol{Y}}(f(x), y') \right)$$

$$\mathtt{fib}_1^{\boldsymbol{f}}(y, y', q) := \mathtt{fib}_{yy'}^{\boldsymbol{f},q} : \sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^{\boldsymbol{Y}}(f(x), y) \to \sum_{x \in \boldsymbol{X}} \mathtt{PrfEq}^{\boldsymbol{Y}}(f(x), y')$$

$$(x, p) \mapsto x, p * q),$$

is a function-like, proof-relevant family of sets over $\boldsymbol{Y}$.

Lemmata 4.8.1. and 4.8.2. in book-HoTT

If $\Lambda := (\lambda_0, \lambda_1)$ is an $\boldsymbol{I}$-family of sets, where $\boldsymbol{I}$ is a Martin-Löf set, then for every $i \in \boldsymbol{I}$

$$\mathtt{fib}^{\mathrm{pr}_1^{\Lambda}}(i) =_{\mathbb{V}_0} \lambda_0(i)$$

As in Book-HoTT, we use the left universal property of identity types

$$\left(\sum_{j \in \boldsymbol{I}} \sum_{p \in \mathtt{PrfEq}^{\boldsymbol{I}}(j,i)} \lambda_0(j) \right) =_{\mathbb{V}_0} \lambda_0(i)$$

If $\boldsymbol{f} : \boldsymbol{X} \to \boldsymbol{Y}$, then

$$X =_{\mathbb{V}_0} \sum_{y \in \boldsymbol{Y}} \mathtt{fib}^{\boldsymbol{f}}(y)$$


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